Database

Introduction to the Concepts of Database Management System (DBMS)

A **Database Management System (DBMS)** is software that helps create, manage, organize, and retrieve data efficiently in a structured way.

Key concepts:

- **Database** → A collection of related data stored in an organized manner (e.g., student records, employee details).
- **Table** → Basic unit; like a spreadsheet with rows (records) and columns (fields).
- **Field** → A single piece of information (column), e.g., Name, Roll No., Date of Birth.
- **Record** → A complete row of data about one entity (e.g., one student's full details).
- **Primary Key** → Unique identifier for each record (e.g., Roll No. or Student ID).
- **Query** → Question asked to the database to retrieve, filter, or calculate data.
- **Form** → User-friendly interface for entering/viewing data (like a data entry screen).
- **Report** → Formatted output for printing or viewing summarized data.

Advantages of DBMS: Data security, reduced redundancy, easy backup, multi-user access, and fast searching.

Here are examples of the Microsoft Access main interface when you first open it (showing blank database creation options):

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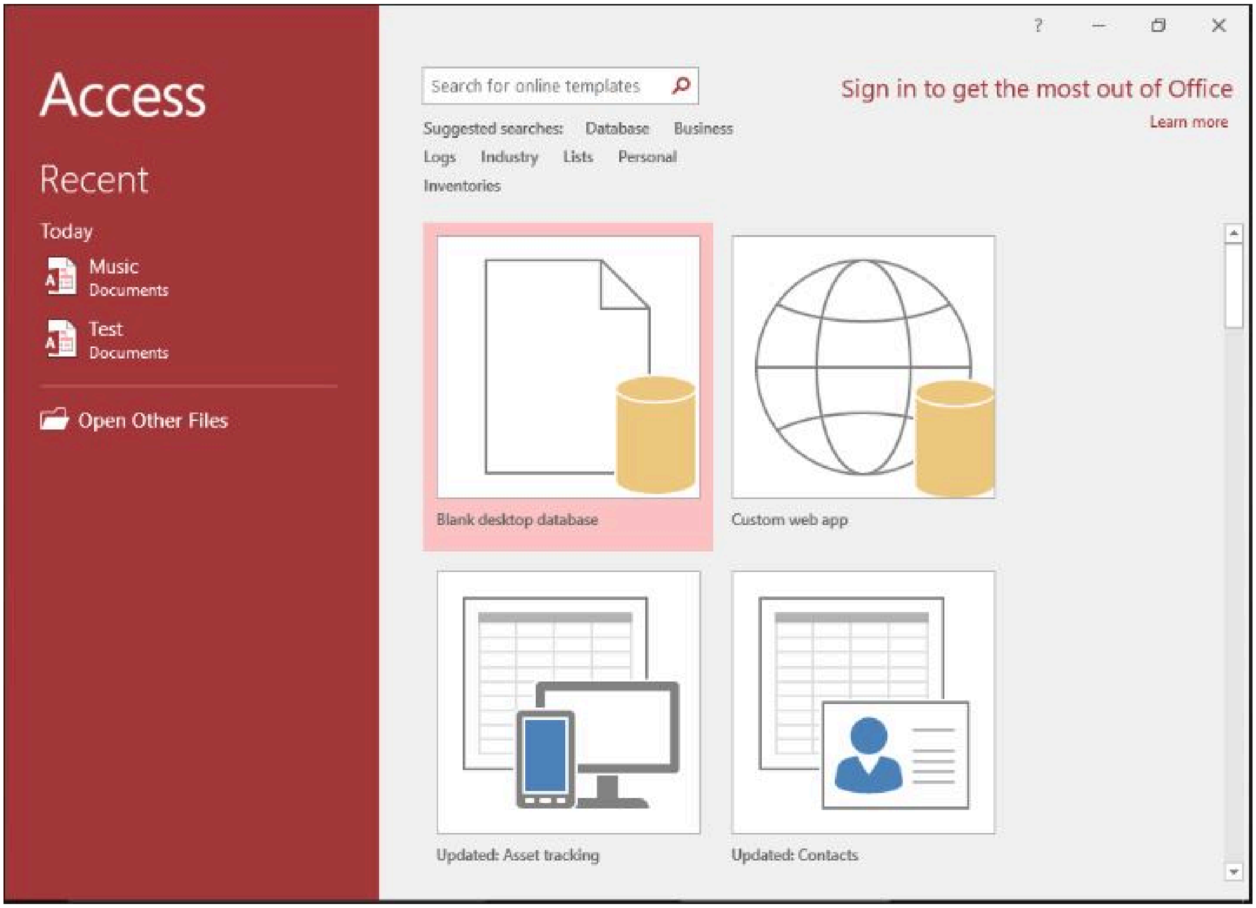
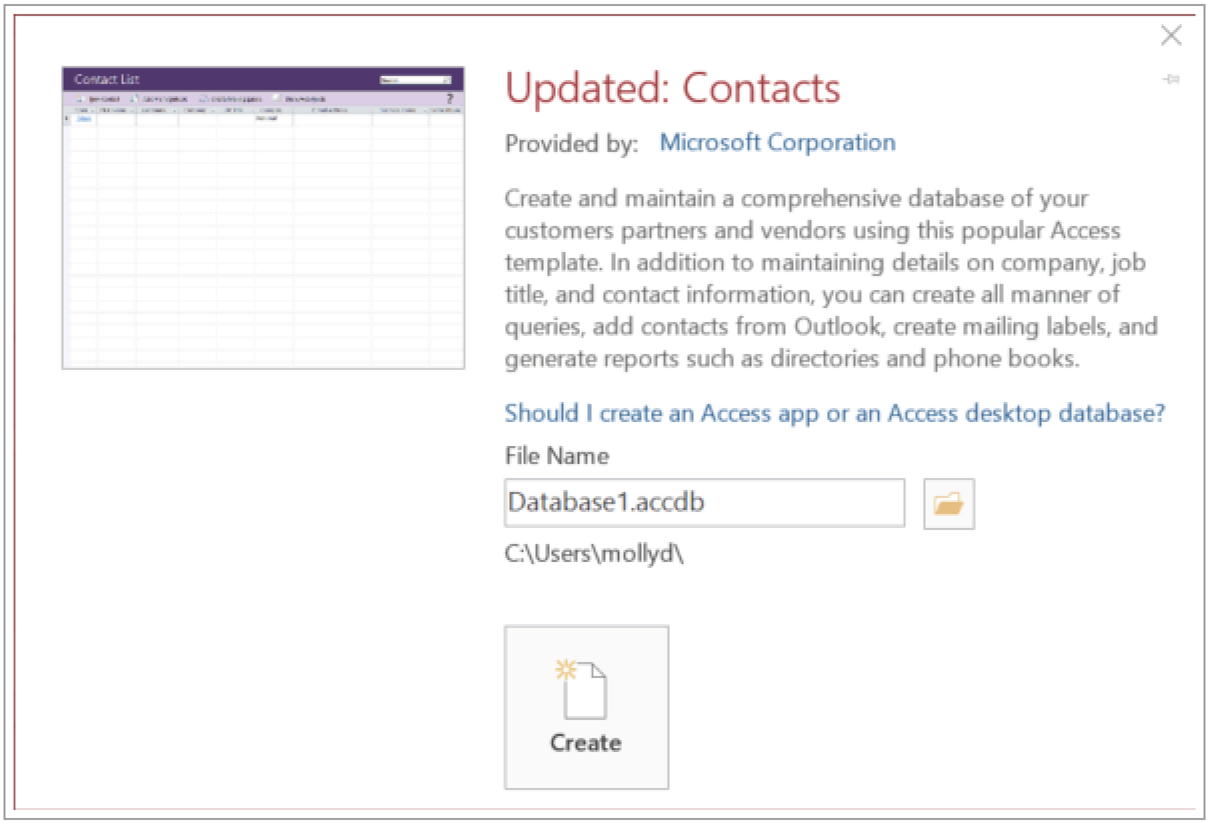


mlpp.pressbooks.pub

Creating a Database

1. Open **Microsoft Access**.
2. On the start screen → **Blank desktop database** (or choose a template).
3. Enter a **File Name** (e.g., SchoolDatabase.accdb).
4. Choose location → Click **Create**.

Here is a visual example of creating a new blank database:



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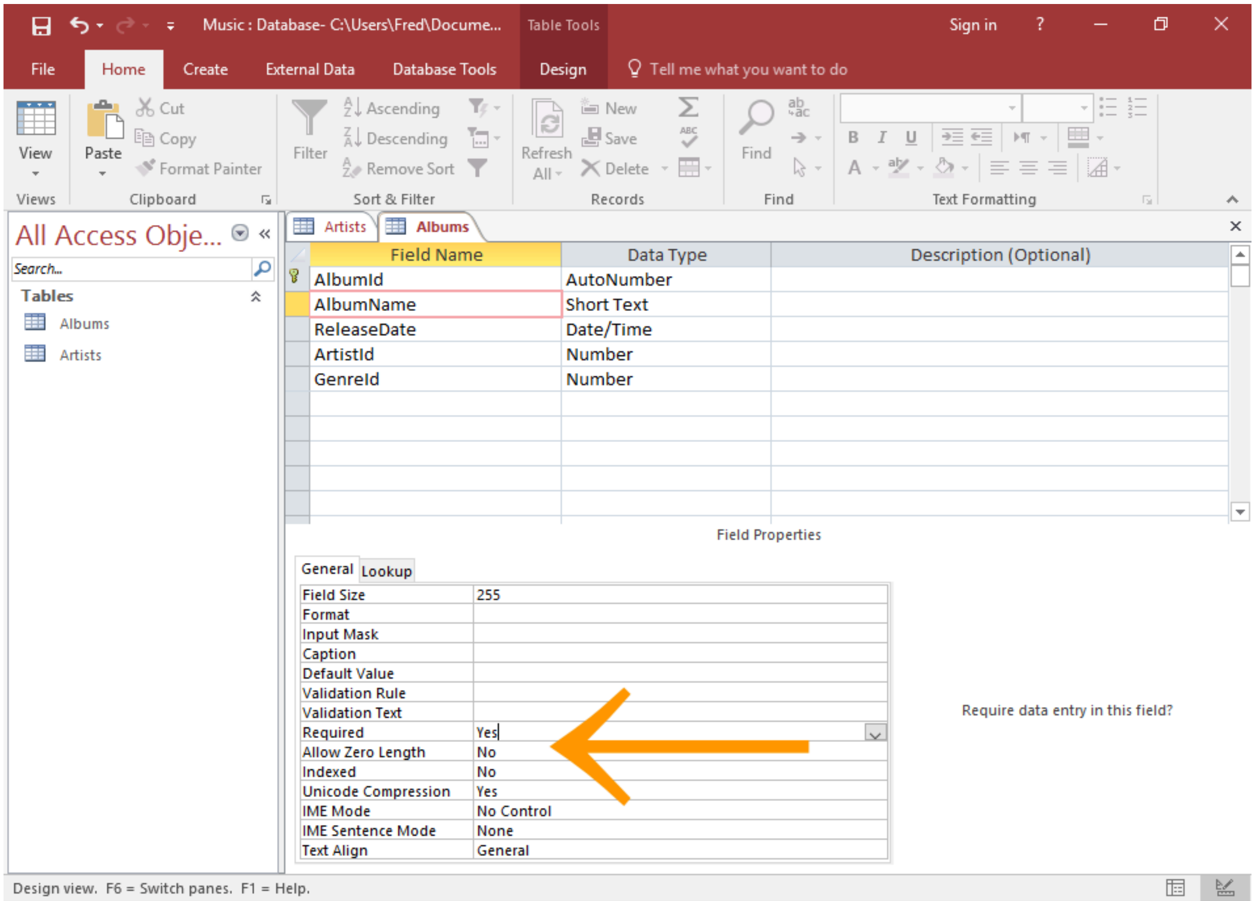
Video: Create an Access desktop database - Microsoft Support

Creating a Table: Concepts of Field, Field Types

Tables are created in two main ways: **Datasheet View** (easy, like Excel) or **Design View** (recommended for control).

- **Field** → Column name (e.g., StudentID, Name).
- **Common Field Types (Data Types)**:
  - Short Text → Names, addresses (up to 255 characters).
  - Long Text → Large descriptions.
  - Number → For calculations (Integer, Double, etc.).
  - Date/Time → Dates and times.
  - Yes/No → Checkbox (True/False).
  - AutoNumber → Automatic unique number (great for Primary Key).

Here are examples of creating a table in **Design View** (showing fields, data types, and primary key setting):



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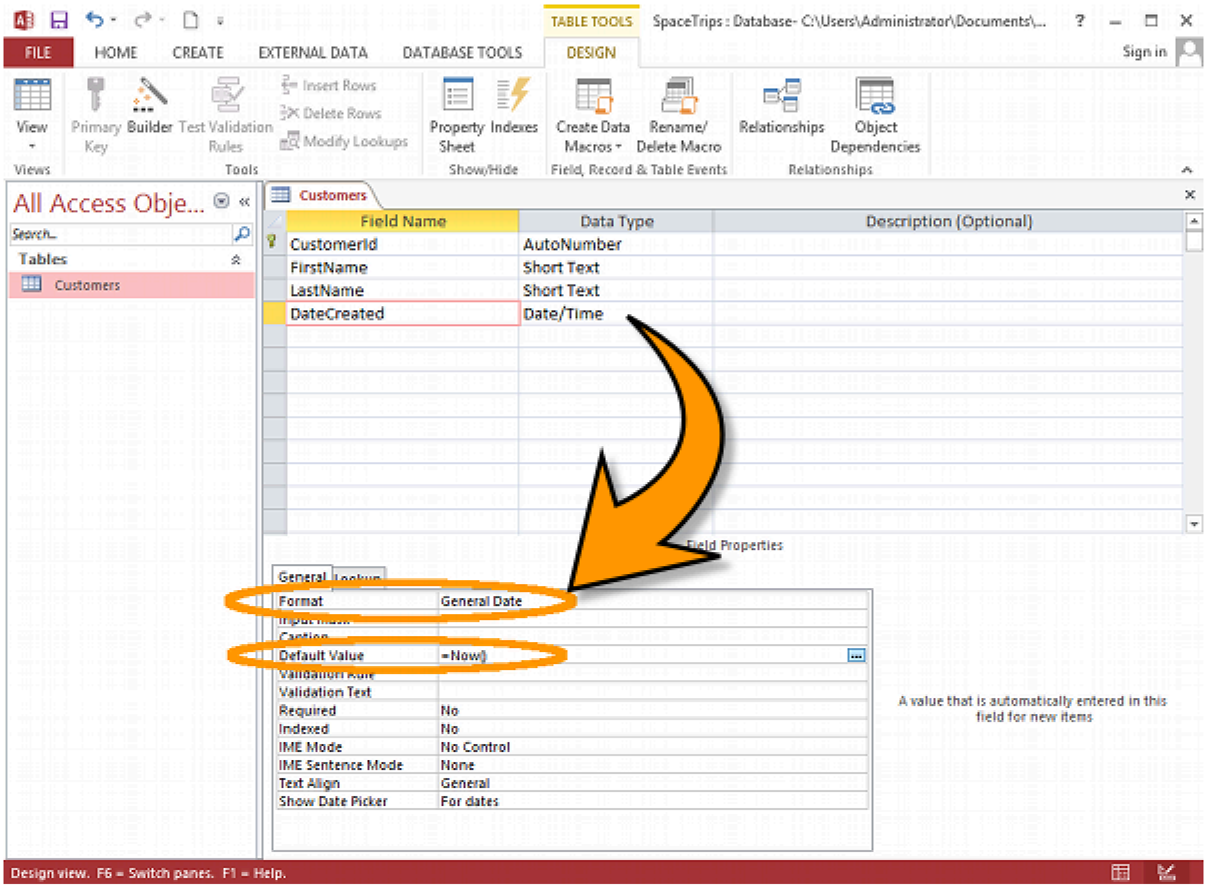

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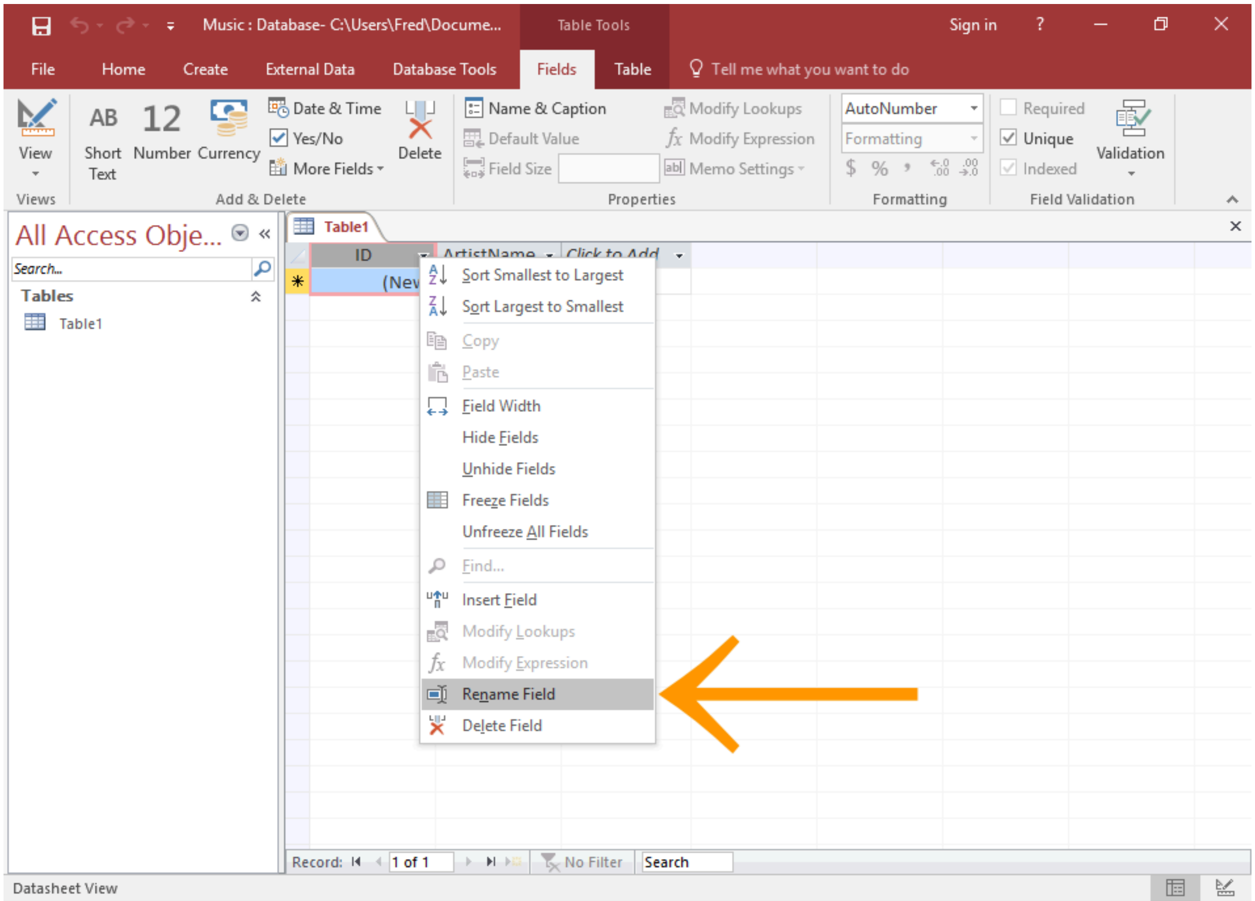
Entering Data in a Table, Preview and Print a Table, Changing Row/Column Height

- Switch to **Datasheet View** → Type data directly (like Excel).
- **Change Row/Column Height** → Drag borders between rows/columns.
- **Preview & Print** → File → Print → Print Preview shows table layout.

Here are examples of entering data in **Datasheet View**:



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How to Create a Table in Datasheet View in Access 2016

Closing and Opening of Table, Sorting of Table, Finding and Replacing Texts

- **Close Table** → Click × on table tab.
- **Open Table** → Double-click table name in Navigation Pane (left side).
- **Sorting** → Click column header → Right-click → Sort A to Z / Z to A.
- **Find/Replace** → Home tab → Find (or Ctrl + F) → Find/Replace tab.

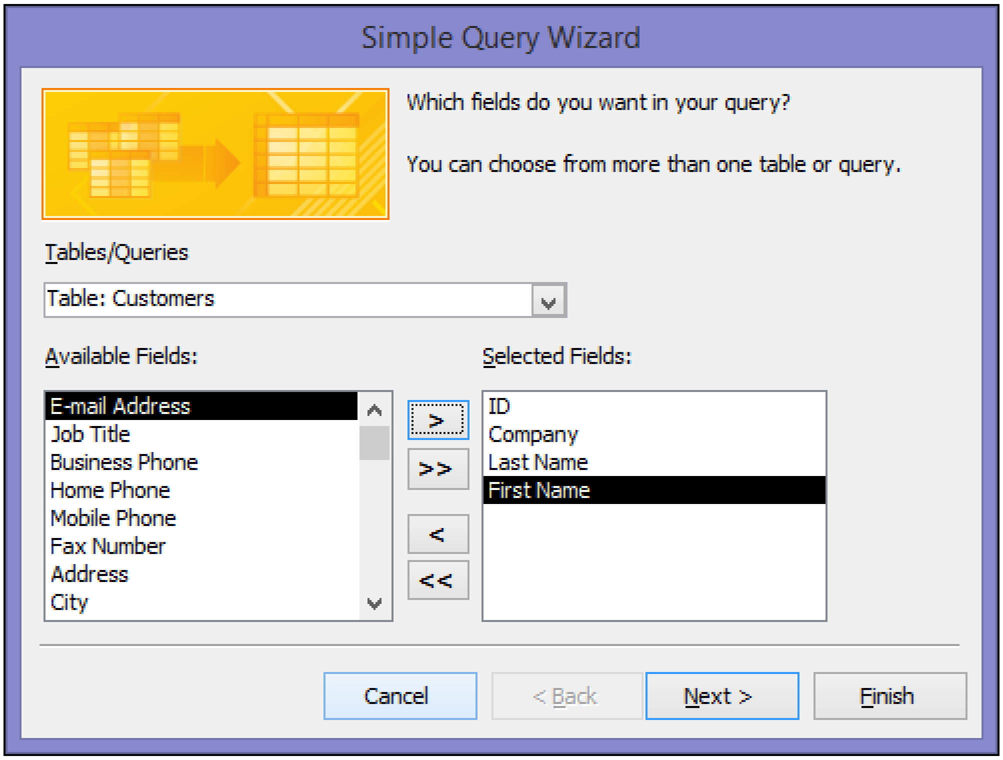
Using Queries Wizard

Queries help filter, sort, and show specific data.

1. Create tab → Query Wizard.

2. Choose **Simple Query Wizard**.
3. Select table → Add fields → Next → Finish.
4. View results in Datasheet View.

Here are step-by-step examples of the **Query Wizard** in action:



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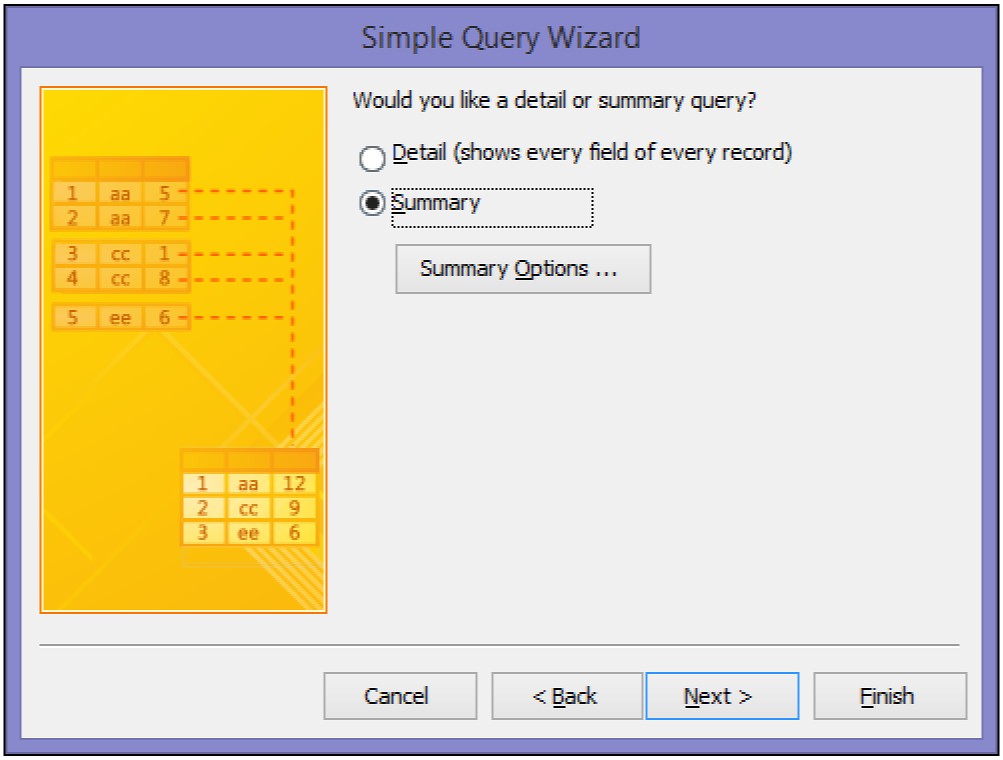

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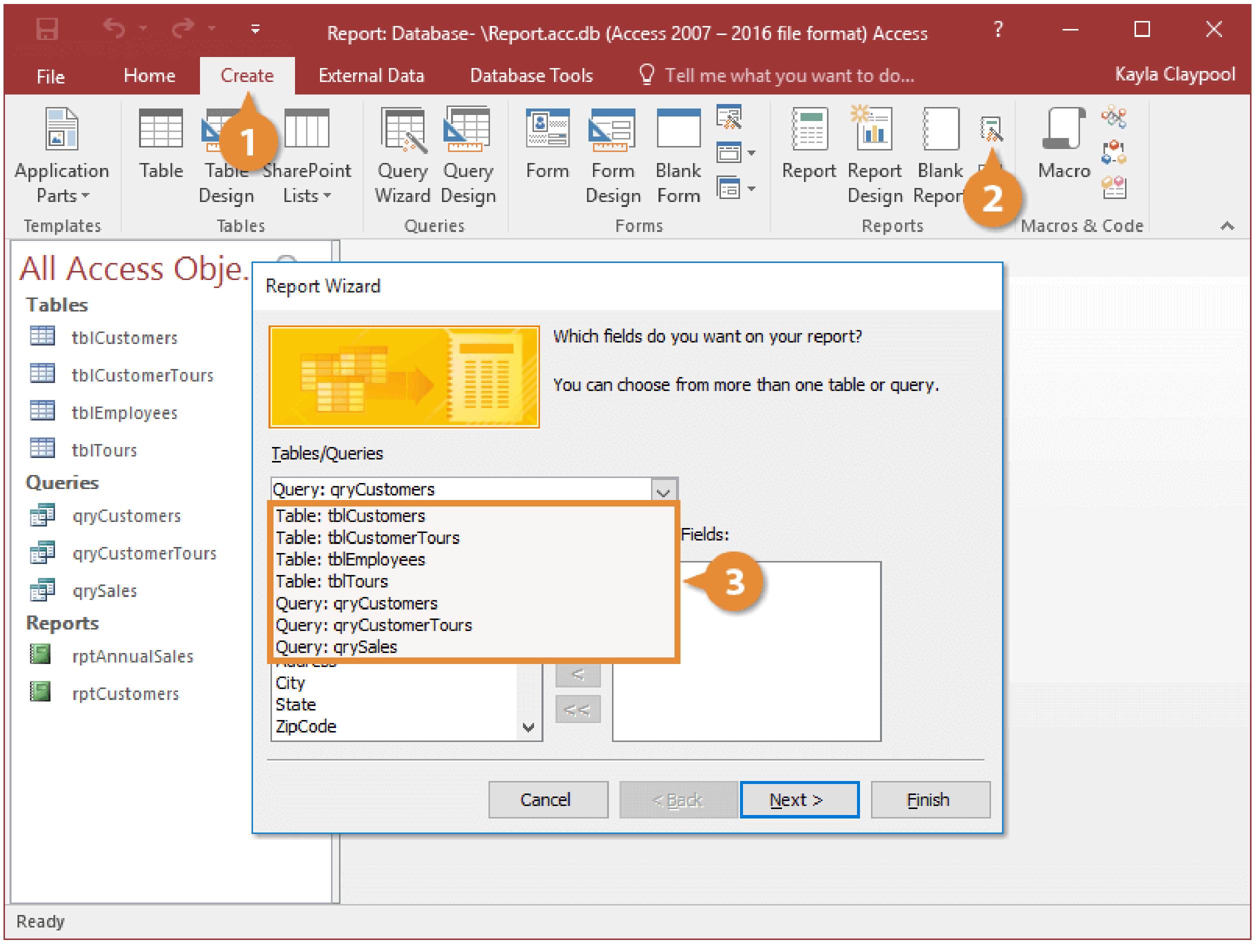
Creating Report from Tables/Queries from Report Wizard, Modifying a Report, Printing of Report

Reports present data nicely for printing.

1. Create tab → Report Wizard.
2. Select table/query → Add fields → Group/Sort options → Layout → Finish.
3. **Modify** → In Layout/Design View → Add labels, change fonts/colors.
4. **Print** → File → Print.

Here are examples of the **Report Wizard** steps and a finished formatted report:



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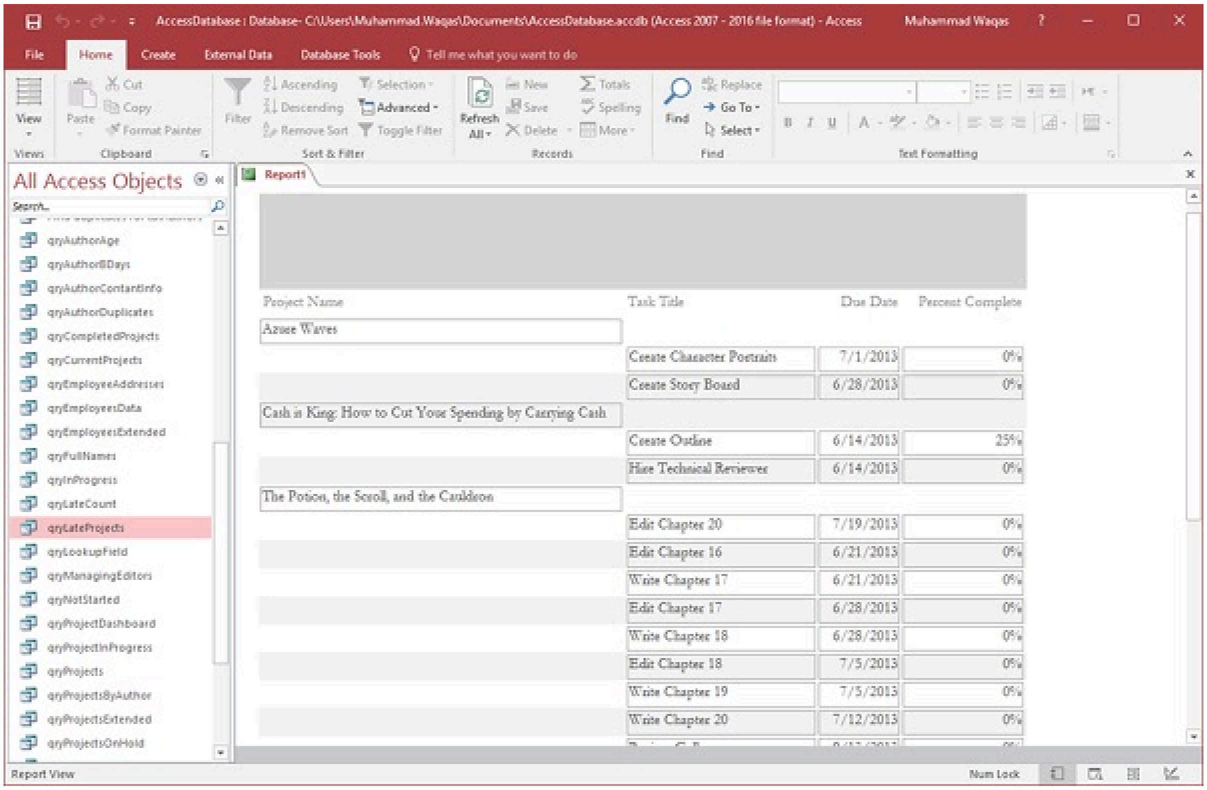

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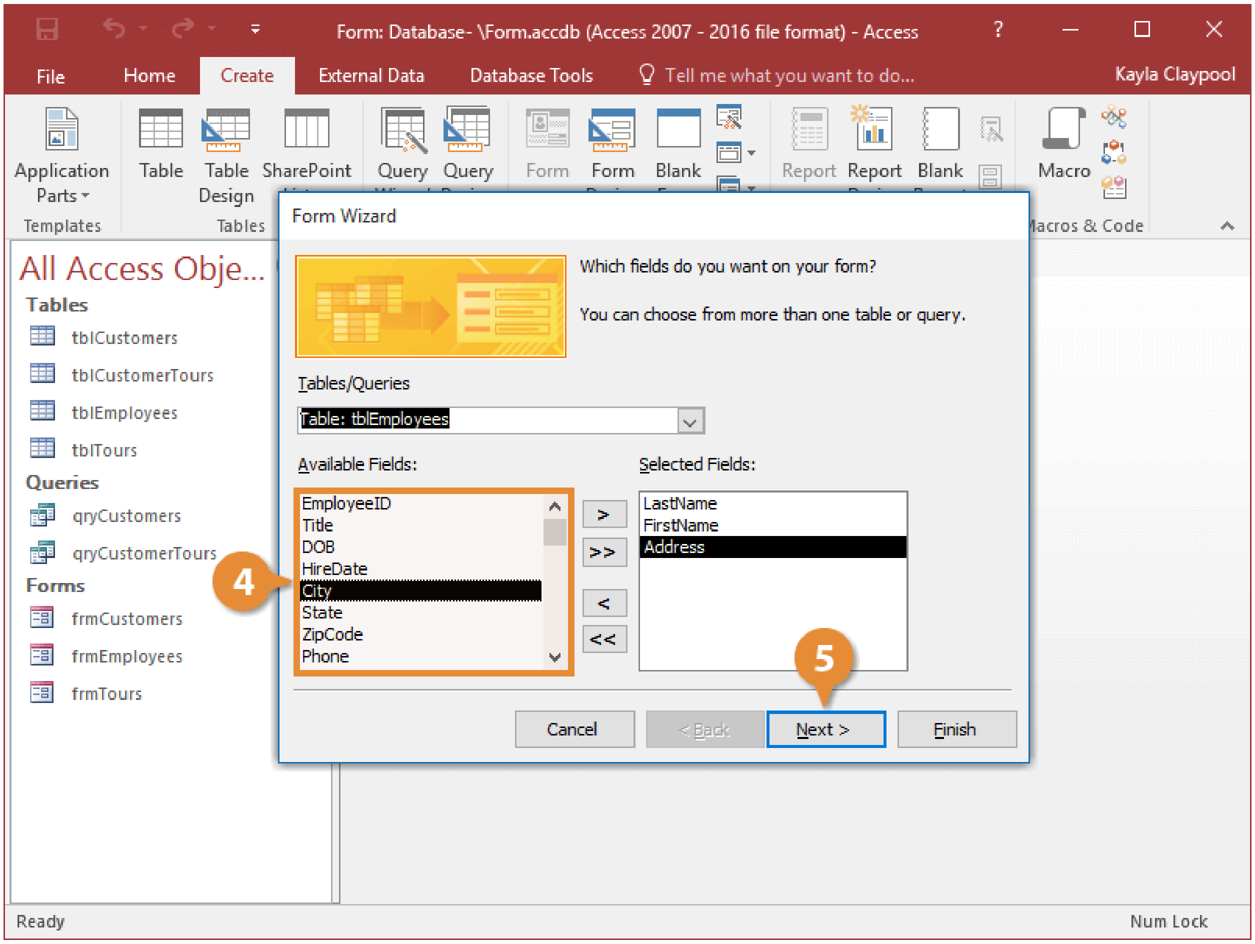
Creating a Form Using Wizard, Entry in the Forms, Basics of Formatting of Forms and Reports

Forms make data entry easy and attractive.

1. Create tab → Form Wizard.
2. Select table/query → Add fields → Layout (Columnar, Tabular, etc.) → Finish.
3. **Entry** → Open form → Use navigation buttons (New Record, Next, Previous).
4. **Formatting** → In Layout/Design View → Change colors, fonts, add logos, adjust sizes.

Here are visual examples of the **Form Wizard** and a created form:



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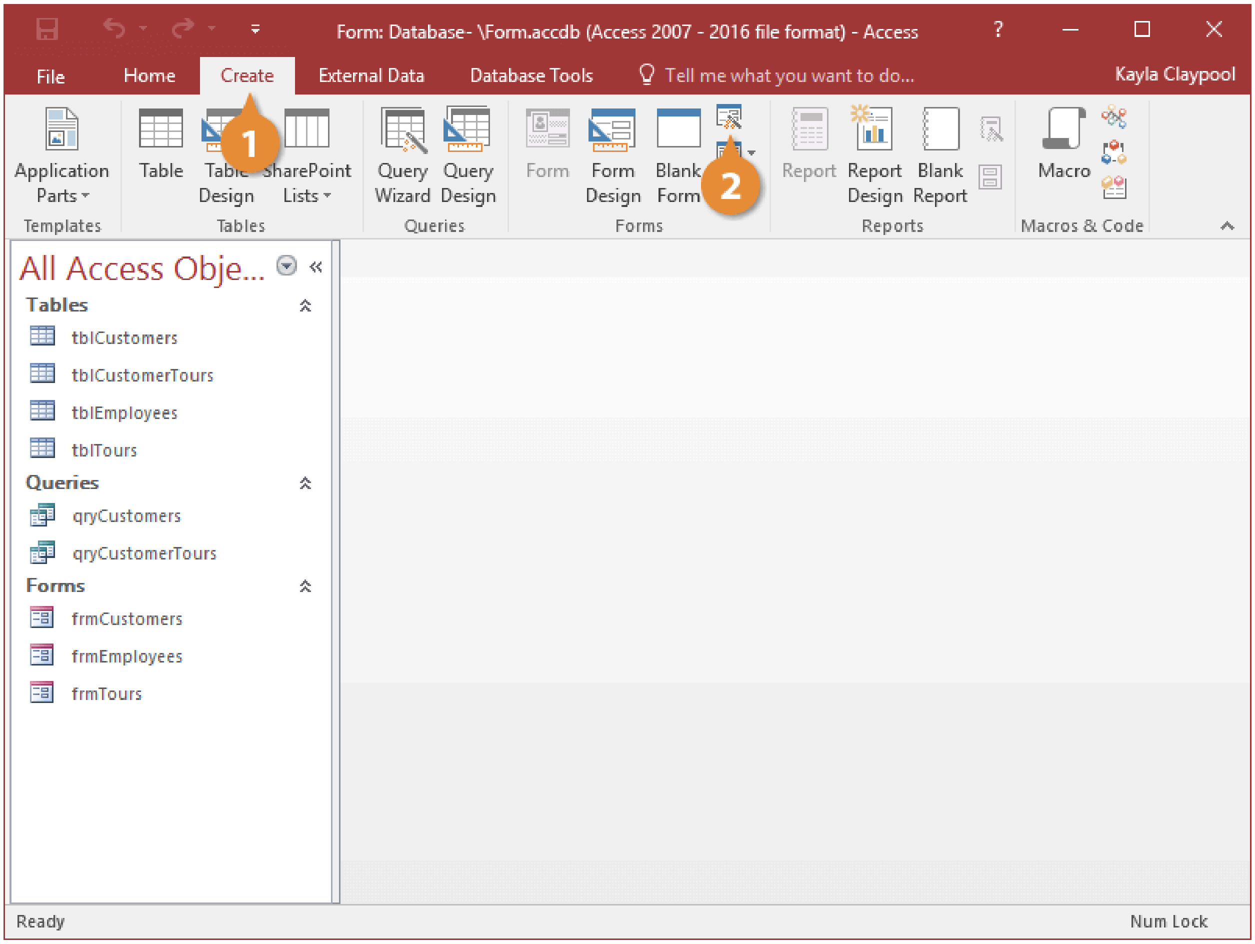

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